

Award Abstract # 1831177**STTR Phase II: A Low Cost Robotics kit for Elementary Education**

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ABSTRACT

This STTR Phase II project supports standards-based math education in elementary school classrooms with a hands-on technology intervention. Research has shown that many elementary teachers suffer from low confidence and limited subject content knowledge in math and struggle to develop instruction designed to meet or exceed common core math learning goals. Teachers and researchers alike seek new approaches to engage students and improve teacher effectiveness to improve learning outcomes. The primary goal of this project is the development of a flexible, user-friendly, hands-on robotics kit with associated curriculum and support for teachers, that will engage students in learning math content, align with core curriculum, and measurably increase student achievement. The commercialization of this research-based classroom kit will enable school districts to adopt

active learning into their math pedagogy. Ultimately, this promotes the NSF mission to increase national prosperity through science innovation by improving math preparation for students across the United States and preparing them to participate in careers that drive the advancement of science and technology.

The core contribution of this work is composed of a flexible hardware kit to enable active learning within the core elementary curriculum as well as more traditional maker activities, and a suite of apps that allow students to use this kit to learn specific math content while also providing options to learn computational thinking through general purpose programming apps. To accomplish this, the team employs a proven design process in which hardware, software, and curriculum are simultaneously designed to align to learner goals, evaluated in classroom studies, and iteratively refined. The kit will combine the ease of use and simplicity of a regular snap-together style electronics kit with the flexibility of a programmable microcontroller. The apps developed for this project will build on a new math-based paradigm for robot programming. These math-oriented apps will remove the barrier of programming skills for elementary teachers and students alike when using the electronics kit for math instruction. Simultaneously, programming apps will enable open-ended explorations of making and computational thinking. Another contribution of this project will be the testing and analysis of the hardware system and complementary math curricula. Formative evaluation will enable exploration and understanding of novel mechanisms for learning math, and evaluation of the program's efficacy will enable characterization of the impact on student outcomes in math achievement and attitudes towards math.

This award reflects NSF's statutory mission and has been deemed worthy of support through evaluation using the Foundation's intellectual merit and broader impacts review criteria.

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Zito, Lauren and Cross, Jennifer L. and Brewer, Bambi and Speer, Samantha and Tasota, Michael and Hamner, Emily and Johnson, Molly and Lauwers, Tom and Nourbakhsh, Illah "Leveraging tangible interfaces in primary school math: Pilot testing of the Owlet math program" *International Journal of Child-Computer Interaction*, v.27, 2021
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PROJECT OUTCOMES REPORT

Disclaimer

This Project Outcomes Report for the General Public is displayed verbatim as submitted by the Principal Investigator (PI) for this award. Any opinions, findings, and conclusions or recommendations expressed in this Report are those of the PI and do not necessarily reflect the views of the National Science Foundation; NSF has not approved or endorsed its content.

The three year design and research study carried out under this award has led to the development of two app-connected manipulatives that allow students in grades K-5 to playfully practice math skills while sparking breakthroughs in math understanding. These manipulatives, dubbed "GlowBoard" and "CubeTower" will be commercialized by BirdBrain Technologies in 2022 under the brand name Owlet.

The Owlet GlowBoard allows students to practice basic arithmetic and fractions using a manipulative consisting of a 12x12 matrix of LED lights. Students complete math exercises using dials and buttons that control the lights. The Owlet CubeTower focuses on understanding place value; students create numbers by placing cubes in the sensor-enabled tracks. The manipulatives connect to a variety of tablet and computer apps. The exercises students solve in these apps are aligned to common core math standards, specifically those covering place value, arithmetic, and fractions. The apps have built-in difficulty levels to allow teachers to select exercises at an appropriate level for each student.

As stated in the abstract, the original focus of the Owlet research project was on integrating math exercises with a novel robotics kit. Although the GlowBoard and CubeTower represent novel educational hardware, they are not recognizably robots. The project's pivot towards app-connected manipulatives is due to frequent consultation with math teachers in the form of interviews, focus groups, and classroom pilots. In the

course of the research, the team worked with 30+ math teachers, and tested products in 16 classrooms with 302 students.

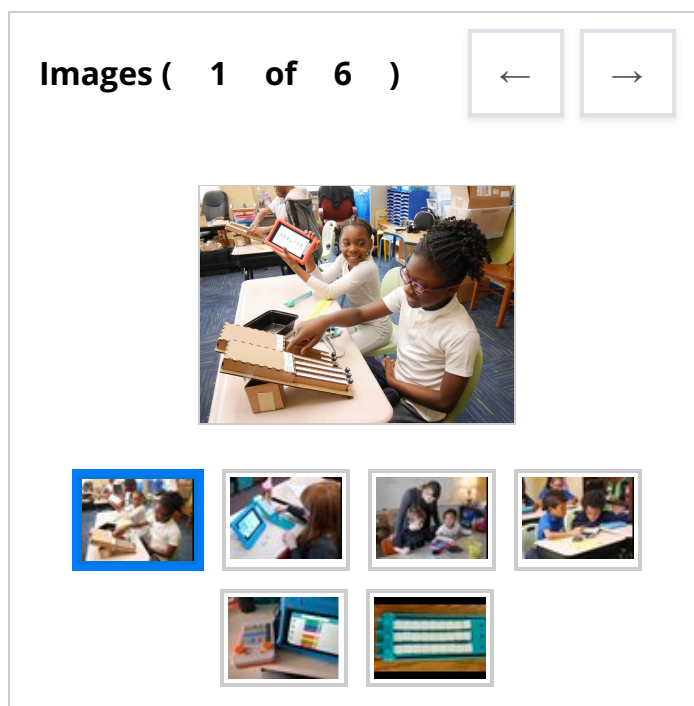
Six teachers participated in a year-long study in which students were provided with both GlowBoard and CubeTower manipulatives and teachers were trained to use these in their math practice. Teachers were interviewed monthly to understand how they were using Owlet, student use of the devices was tracked, and students were tested on measures of math confidence at the start and end of the year. The study occurred in the 2020-21 school year, and so the use of Owlet was studied in unusual educational settings, such as remote and hybrid learning. In-person learning was altered as well, as students were unable to work in pairs in most classrooms due to social distancing policies. Despite these difficulties, a preliminary analysis of student data and teacher interviews suggests that:

- Student engagement with Owlet was high and remained high across the school year.
- Owlet was able to be incorporated into various math curricula across the elementary grades and used in a variety of ways: remote and in-person instruction, small group and large group instruction, individual and partner work, differentiated learning, and student led exploration.
- Students enjoyed using Owlet and would actively choose to use it when given the choice of various math activities.
- Technology-related hurdles were minor and did not outweigh teacher-reported positives of using Owlet.
- Teachers and students enjoyed the hands-on element that Owlet provided.
- Teachers felt that Owlet enhanced student learning and motivation.

The research behind Owlet has directly and positively impacted several hundred students and teachers who participated in the research process, and has led to the development of a novel set of math manipulatives that merge the strengths of tangibles with those of app-based feedback and tracking. If the education marketplace responds well to Owlet, it is likely to have a positive impact on elementary school math learning for many thousands of students.

The work summarized here is the result of a long-standing collaboration between BirdBrain Technologies, an educational electronics and robotics firm founded in 2010, and the Community Robotics Education and Technology Empowerment (CREATE) Lab at Carnegie Mellon University (CMU). BirdBrain Technologies has two existing products, the Finch Robot and the Hummingbird Robotics Kit, both of which resulted from NSF-funded research conducted at the CREATE Lab.

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Modified by: Tom B Lauwers



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